

JEE MAIN REVISION MASTER: SOME BASIC CONCEPTS OF CHEMISTRY SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 40 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Mole Concept & Stoichiometry	33 PYQs	Stoichiometry
High	Concentration Terms (Molarity, Molality, Mole Fraction)	14 PYQs	Concentration

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mole Concept & Stoichiometry	$n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{\text{STP}}}{22.4}$ $\text{Combustion: } \text{C}_x\text{H}_y + \left(x + \frac{y}{4}\right)\text{O}_2 \rightarrow x\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$
Concentration Terms (Molarity, Molality, Mole Fraction)	$M = \frac{w_B \times 1000}{M_B \times V_{\text{mL}}}$, $m = \frac{w_B \times 1000}{M_B \times W_A}$ (g), $x_B = \frac{n_B}{n_A + n_B}$, $\text{ppm} = \frac{w_B}{w_{\text{soln}}} \times 10^6$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mole Concept & Stoichiometry (Very High Yield — 33 PYQs)

Core Principle: Mole relationships, mass-volume stoichiometry, empirical vs molecular formulas, and limiting reagent identification.

■ **Caution / JEE Pitfall:** Always balance the chemical reaction before calculating limiting reagent moles/stoichiometric coefficient ratios.

■ **Speed Trick:** The reactant with the SMALLEST ratio of (Initial Moles / Stoichiometric Coefficient) is the limiting reagent.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II. List - I Mass of substance List - II Number of atoms A. 1.8 mg water I. $2 \times 10^{-4} \times N_{\dots}$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "How many grams of residue is obtained by heating 2.76 g of silver carbonate? (Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol^{-1} ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of CO_2 (g) formed when 3..." [Ans: N/A]

3.2 Concentration Terms (Molarity, Molality, Mole Fraction) (High Yield — 14 PYQs)

Core Principle: Formulas for solution concentrations, temperature dependence, and dilution relation $M_1 V_1 = M_2 V_2$.

■ **Caution / JEE Pitfall:** Molarity (M) and Normality (N) change with temperature (due to volume expansion), while Molality (m) and Mole Fraction (x) are temperature INDEPENDENT.

■ **Speed Trick:** Direct conversion formula: $m = (1000 M / 1000 d - M M_B)$, where d is solution density in g/mL.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of CO₂ (g) formed when 3..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "In the reaction, $2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#2 (MCQ):** "What is the mole fraction of water in 10% by weight (w/w) of aqueous urea solution? [Given: Molar mass of H, O, C and N are 1, 16, 12 and 14 respectively]" **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#3)

An oxide of iron contains 69.9% iron, its empirical formula, is : (Given : Molar mass of Fe and O are 56 and 16 g mol⁻¹ respectively.)

- (A) FeO
- (B) Fe₂O₃ **[Correct]**
- (C) Fe₃O₄
- (D) FeO₃

Correct Answer: B

Solution Analysis: Let the oxide of iron contain 100 g of the compound. Then, Mass of Fe = 69.9 g Mass of O = 100 – 69.9 = 30.1 g Calculate moles of each element Moles of Fe = (69.9 / 56) = 1.248 Moles of O = (30.1 / 16) = 1.881 Find the simplest whole-number ratio Divide each by the smaller number (1.248): Ratio of Fe:O = (1.248 / 1.248) : (1.881 / 1.248) = 1 : 1.51 This ratio is approximately 1 : 1.5 . To remove the fraction, multiply both by 2 : Fe₂O₃ In Fe₂O₃ , Mass of Fe = 2 × 56 = 112 g Mass of O = 3 × 16 = 48 g Total mass = 160 g Percentag